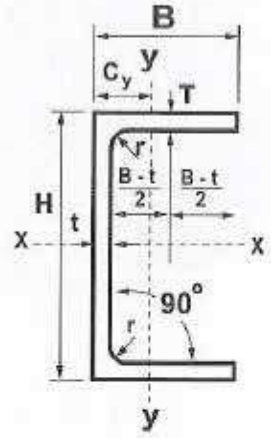
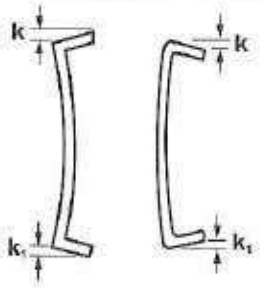
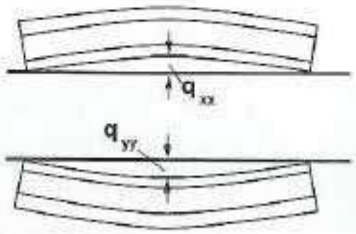
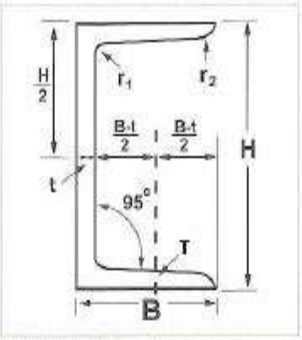
# i) Rolling Tolerances - EN 10279 : 2000
## European Standard

**This European standard specifies requirements for the tolerances on dimensions,shape and mass on hot rolled steel channels with parallel flanges.**

| Property | Range mm | Tolerance mm | | Designation |
|---|---|---|---|---|
| Height **H** | H ≤ 65<br>65 < H ≤ 200<br>200 < H ≤ 400<br>400 < H | ± 1.5<br>± 2.0<br>± 3.0<br>±4.0 | |  |
| Flange width **B** | B ≤ 50<br>50 < B ≤ 100<br>100 < B ≤ 125<br>125 < B | ± 1.5<br>± 2.0<br>± 2.5<br>± 3.0 | | |
| Web thickness **t** | t ≤ 10<br>10 < t ≤ 15<br>15< t | ± 0.5<br>± 0.7<br>± 1.0 | | |
| Flange thickness **T** | T ≤ 10<br>10 < T ≤ 15<br>15 < T | Tolerance limited by weight | -0.5<br>-1<br>-1.5 | |
| Heel radius $r_3$ | All Sizes | < 0.3 * t | | |
| Out of squareness **(k,$k_1$)** | B ≤ 100<br>100 < B | 2<br>2% of B | |  |
| Web flatness **f** | H ≤ 100<br>100 < H ≤ 200<br>200 < H ≤ 400<br>400 < H | ± 0.5<br>± 1.0<br>± 1.5<br>± 1.5 | | |
| Straightness **qxx** | H ≤ 150<br>150 < H ≤ 300<br>300 < H | 0.30% L<br>0.20% L<br>0.15% L | |  |
| Straightness **qyy** | H ≤ 150<br>150 < H ≤300<br>300 < H | 0.50% L<br>0.30% L<br>0.20% L | | |
| Mass per unit length kg/m | H ≤ 125<br>125 < H | ± 6.0%<br>± 4.0% | | |
| Standard Length | All | 100 | 0 | |
| Alternative Standard (by agreement) **L** | All | 50 | -50 | |

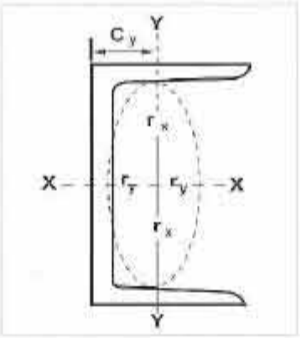
## Dimensions and Properties




# 5a(i) U Channels (Imperial Units)- Part 1

| Designation Size | Unit Mass | | Depth Of Section | | Width Of Section | | Thickness | | | | Root Radius | | Toe Radius | | Area Of Section | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Web | | Flange | | | | | | | |
| H x B | M | | H | | B | | t | | T | | $r_1$ | | $r_2$ | | A | |
| mm (in) | kg/m | lb/ft | mm | in | mm | in | mm | in | mm | in | mm | in | mm | in | $cm^2$ | $in^2$ |
| 127 x 64 (5 x 2 1/2) | 14.90 | 10.0 | 127.0 | 5.0 | 63.5 | 2.5 | 6.4 | 0.252 | 9.2 | 0.362 | 10.7 | 0.421 | 2.4 | 0.094 | 18.98 | 2.94 |
| 152 x 76 (6 x 3) | 17.88 | 12.0 | 152.4 | 6.0 | 76.2 | 3.0 | 6.4 | 0.252 | 9.0 | 0.354 | 12.2 | 0.480 | 2.4 | 0.094 | 22.77 | 3.53 |
| 152 x 89 (6 x 3 1/2) | 23.84 | 16.0 | 152.4 | 6.0 | 88.9 | 3.5 | 7.1 | 0.280 | 11.6 | 0.457 | 13.7 | 0.539 | 3.2 | 0.126 | 30.36 | 4.71 |
| 178 x 76 (7 x 3) | 20.84 | 14.0 | 177.8 | 7.0 | 76.2 | 3.0 | 6.6 | 0.260 | 10.3 | 0.406 | 12.2 | 0.480 | 3.2 | 0.126 | 26.54 | 4.11 |
| 178 x 89 (7 x 3 1/2) | 26.81 | 18.0 | 177.8 | 7.0 | 88.9 | 3.5 | 7.6 | 0.299 | 12.3 | 0.484 | 13.7 | 0.539 | 3.2 | 0.126 | 34.15 | 5.29 |
| 203 x 76 (8 x 3) | 23.82 | 16.0 | 203.2 | 8.0 | 76.2 | 3.0 | 7.1 | 0.280 | 11.2 | 0.441 | 12.2 | 0.480 | 3.2 | 0.126 | 30.34 | 4.70 |
| 203 x 89 (8 x 3 1/2) | 29.78 | 20.0 | 203.2 | 8.0 | 88.9 | 3.5 | 8.1 | 0.319 | 12.9 | 0.508 | 13.7 | 0.539 | 3.2 | 0.126 | 37.94 | 5.88 |
| 229 x 76 (9 x 3) | 26.06 | 17.5 | 228.6 | 9.0 | 76.2 | 3.0 | 7.6 | 0.299 | 11.2 | 0.441 | 12.2 | 0.480 | 3.2 | 0.126 | 33.20 | 5.15 |
| 229 x 89 (9 x 3 1/2) | 32.76 | 22.0 | 228.6 | 9.0 | 88.9 | 3.5 | 8.6 | 0.339 | 13.3 | 0.524 | 13.7 | 0.539 | 3.2 | 0.126 | 41.73 | 6.47 |
| 254 x 76 (10 x 3) | 28.29 | 19.0 | 254.0 | 10.0 | 76.2 | 3.0 | 8.1 | 0.319 | 10.9 | 0.429 | 12.2 | 0.480 | 3.2 | 0.126 | 36.03 | 5.58 |
| 254 x 89 (10 x 3 1/2) | 35.74 | 24.0 | 254.0 | 10.0 | 88.9 | 3.5 | 9.1 | 0.358 | 13.6 | 0.535 | 13.7 | 0.539 | 3.2 | 0.126 | 45.52 | 7.06 |
| 305 x 89 (12 x 3 1/2) | 41.69 | 28.0 | 304.8 | 12.0 | 88.9 | 3.5 | 10.2 | 0.402 | 13.7 | 0.539 | 13.7 | 0.539 | 3.2 | 0.126 | 53.11 | 8.23 |
| 305 x 102 (12 x 4) | 46.18 | 31.0 | 304.8 | 12.0 | 101.6 | 4.0 | 10.2 | 0.402 | 14.8 | 0.583 | 15.2 | 0.598 | 4.8 | 0.189 | 58.83 | 9.12 |
| 381 x 102 (15 x 4) | 55.10 | 37.0 | 381.0 | 15.0 | 101.6 | 4.0 | 10.4 | 0.409 | 16.3 | 0.642 | 15.2 | 0.598 | 4.8 | 0.189 | 70.19 | 10.88 |
| 432 x 102 (17 x 4) | 65.54 | 44.0 | 431.8 | 17.0 | 101.6 | 4.0 | 12.2 | 0.480 | 16.8 | 0.661 | 15.2 | 0.598 | 4.8 | 0.189 | 83.49 | 12.94 |

Note: Calculated based on 1 kg=2.2046 lb, 1 m=3.2808 feet and 1 in = 2.54 cm




## 5a (i) U Channels (Imperial Units)- Part 2

| Moment Of Inertia | | | | Radius Of Gyration | | | | Elastic Modulus | | | | Plastic Modulus | | | | Centre of Gravity | | Designation Size |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Axis x-x | | Axis y-y | | Axis x-x | | Axis y-y | | Axis x-x | | Axis y-y | | Axis x-x | | Axis y-y | | | | |
| $I_x$ | | $I_y$ | | $r_x$ | | $r_y$ | | Zx | | Zy | | Sx | | Sy | | Cy | | H x B |
| $cm^4$ | $in^4$ | $cm^4$ | $in^4$ | cm | in | cm | in | $cm^3$ | $in^3$ | $cm^3$ | $in^3$ | $cm^3$ | $in^3$ | $cm^3$ | $in^3$ | cm | in | mm (in) |
| 482.5 | 11.59 | 67.23 | 1.62 | 5.04 | 1.984 | 1.88 | 0.740 | 75.99 | 4.637 | 15.25 | 0.931 | 89.4 | 5.456 | 29.31 | 1.789 | 1.94 | 0.764 | 127 x 64 (5 x 2 1/2) |
| 851.5 | 20.46 | 113.8 | 2.73 | 6.12 | 2.409 | 2.24 | 0.882 | 111.8 | 6.822 | 21.05 | 1.285 | 130.0 | 7.933 | 41.26 | 2.518 | 2.21 | 0.870 | 152 x 76 (6 x 3) |
| 1166 | 28.01 | 215.1 | 5.17 | 6.20 | 2.441 | 2.66 | 1.047 | 153.0 | 9.337 | 35.70 | 2.178 | 177.7 | 10.844 | 68.12 | 4.157 | 2.86 | 1.126 | 152 x 89 (6 x 3 1/2) |
| 1337 | 32.12 | 134.0 | 3.22 | 7.10 | 2.795 | 2.25 | 0.886 | 150.4 | 9.178 | 24.72 | 1.509 | 175.4 | 10.704 | 48.07 | 2.933 | 2.20 | 0.866 | 178 x 76 (7 x 3) |
| 1753 | 42.12 | 241.0 | 5.79 | 7.16 | 2.819 | 2.66 | 1.047 | 197.2 | 12.034 | 39.29 | 2.398 | 229.6 | 14.011 | 75.44 | 4.604 | 2.76 | 1.087 | 178 x 89 (7 x 3 1/2) |
| 1950 | 46.85 | 151.3 | 3.63 | 8.02 | 3.157 | 2.23 | 0.878 | 192.0 | 11.717 | 27.59 | 1.684 | 225.2 | 13.743 | 53.32 | 3.254 | 2.13 | 0.839 | 203 x 76 (8 x 3) |
| 2491 | 59.85 | 264.4 | 6.35 | 8.10 | 3.189 | 2.64 | 1.039 | 245.2 | 14.963 | 42.34 | 2.584 | 286.6 | 17.489 | 81.62 | 4.981 | 2.65 | 1.043 | 203 x 89 (8 x 3 1/2) |
| 2610 | 62.70 | 158.7 | 3.81 | 8.87 | 3.492 | 2.19 | 0.862 | 228.3 | 13.932 | 28.22 | 1.722 | 270.3 | 16.495 | 54.24 | 3.310 | 2.00 | 0.787 | 229 x 76 (9 x 3) |
| 3387 | 81.37 | 285.0 | 6.85 | 9.01 | 3.547 | 2.61 | 1.028 | 296.4 | 18.087 | 44.82 | 2.735 | 348.4 | 21.261 | 86.38 | 5.271 | 2.53 | 0.996 | 229 x 89 (9 x 3 1/2) |
| 3367 | 80.89 | 162.6 | 3.91 | 9.67 | 3.807 | 2.12 | 0.835 | 265.1 | 16.177 | 28.21 | 1.721 | 317.4 | 19.369 | 54.14 | 3.304 | 1.86 | 0.732 | 254 x 76 (10 x 3) |
| 4448 | 106.86 | 302.4 | 7.27 | 9.88 | 3.890 | 2.58 | 1.016 | 350.2 | 21.371 | 46.70 | 2.850 | 414.4 | 25.288 | 89.56 | 5.465 | 2.42 | 0.953 | 254 x 89 (10 x 3 1/2) |
| 7061 | 169.64 | 325.4 | 7.82 | 11.5 | 4.528 | 2.48 | 0.976 | 463.3 | 28.272 | 48.49 | 2.959 | 557.1 | 33.996 | 92.60 | 5.651 | 2.18 | 0.858 | 305 x 89 (12 x 3 1/2) |
| 8214 | 197.34 | 499.5 | 12.00 | 11.8 | 4.646 | 2.91 | 1.146 | 539.0 | 32.892 | 66.59 | 4.064 | 638.3 | 38.951 | 128.1 | 7.817 | 2.66 | 1.047 | 305 x 102 (12 x 4) |
| 14894 | 357.82 | 579.7 | 13.93 | 14.6 | 5.748 | 2.87 | 1.130 | 781.8 | 47.708 | 75.86 | 4.629 | 932.7 | 56.917 | 144.4 | 8.812 | 2.52 | 0.992 | 381 x 102 (15 x 4) |
| 21399 | 514.11 | 628.6 | 15.10 | 16.0 | 6.299 | 2.74 | 1.079 | 991.1 | 60.480 | 80.14 | 4.890 | 1207 | 73.656 | 153.1 | 9.343 | 2.32 | 0.913 | 432 x 102 (17 x 4) |

Note: Calculated based on 1 kg=2.2046 lb, 1 m=3.2808 feet and 1 in = 2.54 cm

# 5) Light Gauge Steels for General Structure

**Japanese Industrial Standard (Extracts from JIS G 3350 : 2005)**

**General Information**

This Japanese Industrial Standard specifies the cold formed light gauge steels for buildings and other structures, (hereafter referred to as "light gauge steels").

The light gauge steels shall be classified into one class, and its symbol, designation according to sectional shape and mark of sectional shape shall be as given in Table 1.

Table 1. Symbol Of Class, Designation According To Sectional Shape And Mark Of Sectional Shape

| Symbol Of Class | Designation According To Sectional Shape | Mark Of Sectional Shape |
|---|---|---|
| SSC 400 | Light channel steel | ⊏ |
| | Lip channel steel | ⊏ |

**Chemical Composition**

The light gauge steel shall be tested in accordance with and its ladle analysis shall be as given in Table 2.

Table 2. Chemical Composition — Unit : %

| Symbol Of Class | C | P | S |
|---|---|---|---|
| SSC 400 | 0.25 max. | 0.050 max. | 0.050 max. |

**Mechanical Strength (Tensile Test)**

The light gauge steel shall be tested in accordance with and its yield point, tensile strength, and elongation shall be as given in Table 3.

Table 3. Mechanical Properties

| Symbol Of Class | Yield Point ($N/mm^2$) | Tensile Strength ($N/mm^2$) | Elongation | | |
|---|---|---|---|---|---|
| | | | Thickness (mm) | Test Piece | % |
| SSC 400 | 245 min. | 400 to 540 | 5 or under | No. 5 | 21 min. |
| | | | Over 5 | No. 1A | 17 min. |

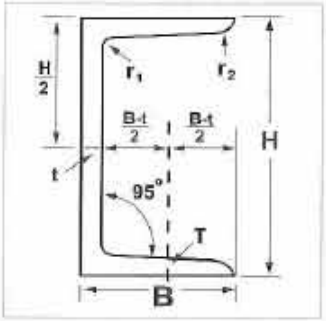
Remark : 1 $N/mm^2$ = 1 MPa

**Tolerances On Dimensions and Mass**

The shape and the tolerance on dimensions of light gauge steel shall be as given in Table 4.

Table 4. Shape And Tolerance On Dimensions

| Division | | Tolerance |
|---|---|---|
| Height ( H ) | Under 150 mm | ± 1.5 mm |
| | 150 mm or over to and excl. 300 mm | ± 2.0 mm |
| | 300 mm or over | ± 3.0 mm |
| Side ( A or B ) | | ± 1.5 mm |
| Lip ( C ) | | ± 2.0 mm |
| Angle between adjacent plate parts | | ± 1.5° |
| Radius of Corner | | ± 1.5 t |
| Length | 7 m or under | + 40 mm<br>- 0 mm |
| | Over 7 m | For each increase of 1m or its fraction in length, add 5 |
| Curvature | | Within 0.2 % of overall length |
| Thickness of plate part | 1.6 mm | ± 0.22 mm |
| | 2.0 mm, 2.3 mm | ± 0.25 mm |
| | 2.8 mm | ± 0.28 mm |
| | 3.2 mm | ± 0.30 mm |
| | 4.0 mm, 4.5 mm | ± 0.45 mm |
| | 6.0 mm | ± 0.60 mm |
| Calculated mass of one set of steel | Under 600 kg | ± 10 % |
| | 600 kg or over to and excl. 2 t | ± 7.5 % |
| | 2 t or over | ± 5 % |

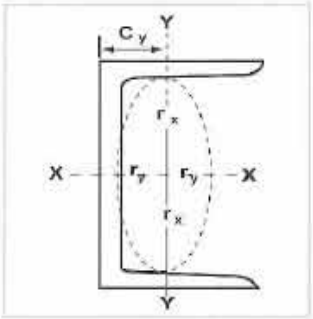



## 5a (ii) U Channels (Metric Units) - Part A

| Designation Size | Dept Of Section | | Width of Flange | | Thickness | | | | Root Radius | | Toe Radius | | Section Area | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Web | | Flange | | | | | | | |
| H x B x t | H | | B | | t | | T | | $r_1$ | | $r_2$ | | A | |
| mm x mm x mm | mm | in | mm | in | mm | in | mm | in | mm | in | mm | in | $cm^2$ | $in^2$ |
| 75 x 40 x 5 | 75 | 2.953 | 40 | 1.575 | 5 | 0.197 | 7 | 0.276 | 8 | 0.315 | 4 | 0.157 | 8.818 | 1.367 |
| 100 x 50 x 5 | 100 | 3.937 | 50 | 1.969 | 5 | 0.197 | 7.5 | 0.295 | 8 | 0.315 | 4 | 0.157 | 11.92 | 1.848 |
| 125 x 65 x 6 | 125 | 4.921 | 65 | 2.559 | 6 | 0.236 | 8 | 0.315 | 8 | 0.315 | 4 | 0.157 | 17.11 | 2.652 |
| 150 x 75 x 6.5 | 150 | 5.906 | 75 | 2.953 | 6.5 | 0.256 | 10 | 0.394 | 10 | 0.394 | 5 | 0.197 | 23.71 | 3.675 |
| 150 x 75 x 9 | 150 | 5.906 | 75 | 2.953 | 9 | 0.354 | 12.5 | 0.492 | 15 | 0.591 | 7.5 | 0.295 | 30.59 | 4.741 |
| 180 x 75 x 7 | 180 | 7.087 | 75 | 2.953 | 7 | 0.276 | 10.5 | 0.413 | 11 | 0.433 | 5.5 | 0.217 | 27.20 | 4.316 |
| 180 x 90 x 7.5 | 180 | 7.087 | 90 | 3.543 | 7.5 | 0.295 | 12.5 | 0.492 | 13 | 0.512 | 6.5 | 0.256 | 34.57 | 5.358 |
| 200 x 80 x 7.5 | 200 | 7.874 | 80 | 3.150 | 7.5 | 0.295 | 11 | 0.433 | 12 | 0.472 | 6 | 0.236 | 31.33 | 4.856 |
| 200 x 90 x 8 | 200 | 7.874 | 90 | 3.543 | 8 | 0.315 | 13.5 | 0.531 | 14 | 0.551 | 7 | 0.276 | 38.65 | 5.991 |
| 230 x 80 x 8 | 230 | 9.055 | 80 | 3.150 | 8 | 0.315 | 12 | 0.472 | 13 | 0.512 | 6.5 | 0.256 | 36.12 | 5.499 |
| 230 x 90 x 8.5 | 230 | 9.055 | 90 | 3.543 | 8.5 | 0.335 | 13.5 | 0.531 | 15 | 0.591 | 7.5 | 0.295 | 42.14 | 6.531 |
| 250 x 80 x 8 | 250 | 9.843 | 80 | 3.150 | 8 | 0.315 | 12.5 | 0.492 | 14 | 0.551 | 7 | 0.276 | 38.51 | 5.969 |
| 250 x 90 x 9 | 250 | 9.843 | 90 | 3.543 | 9 | 0.354 | 13 | 0.512 | 14 | 0.551 | 7 | 0.276 | 44.07 | 6.831 |
| 250 x 90 x 11 | 250 | 9.843 | 90 | 3.543 | 11 | 0.433 | 14.5 | 0.571 | 17 | 0.669 | 8.5 | 0.335 | 51.17 | 7.931 |
| 280 x 100 x 9 | 280 | 11.024 | 100 | 3.937 | 9 | 0.354 | 13 | 0.512 | 14 | 0.551 | 7 | 0.276 | 49.37 | 7.652 |
| 280 x 100 x 11.5 | 280 | 11.024 | 100 | 3.937 | 11.5 | 0.453 | 16 | 0.630 | 18 | 0.709 | 9 | 0.354 | 61.37 | 9.512 |
| 300 x 90 x 9 | 300 | 11.811 | 90 | 3.543 | 9 | 0.354 | 12 | 0.512 | 14 | 0.551 | 7 | 0.276 | 48.57 | 7.531 |
| 300 x 90 x 10 | 300 | 11.811 | 90 | 3.543 | 10 | 0.394 | 15.5 | 0.611 | 19 | 0.748 | 9.5 | 0.374 | 55.74 | 8.640 |
| 380 x 100 x 10.5 | 380 | 14.961 | 100 | 3.937 | 10.5 | 0.431 | 16 | 0.630 | 18 | 0.709 | 9 | 0.354 | 69.39 | 10.76 |
| 380 x 100 x 13 | 380 | 14.961 | 100 | 3.937 | 13 | 0.512 | 16.5 | 0.650 | 18 | 0.709 | 9 | 0.354 | 78.96 | 12.54 |

Note: Calculated based on 1 kg=2.2046 lb, 1 m=3.2808 feet and 1 in = 2.54 cm





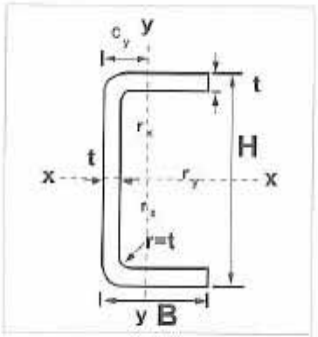

## 5a (ii) U Channels (Metric Units) - Part B

| Unit Mass | | | Centre Of Gravity | | Moment Of Inertial | | | | Radius Of Gyration | | | | Modulus Of Section | | | | Designation Size |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| M | | | Cy | | $I_x$ | | $I_y$ | | $r_x$ | | $r_y$ | | Zx | | Zy | | H x B x t |
| kg/m | kg/ft | lb/ft | cm | in | $cm^4$ | $in^4$ | $cm^4$ | $in^4$ | cm | in | cm | in | $cm^3$ | $in^3$ | $cm^3$ | $in^3$ | mm x mm x mm |
| 6.92 | 2.11 | 4.65 | 1.27 | 0.500 | 75.9 | 1.823 | 12.4 | 0.298 | 2.93 | 1.154 | 1.19 | 0.469 | 20.2 | 1.233 | 4.54 | 0.277 | 75 x 40 x 5 |
| 9.36 | 2.85 | 6.29 | 1.55 | 0.610 | 189 | 4.541 | 26.9 | 0.646 | 3.98 | 1.567 | 1.50 | 0.591 | 37.8 | 2.307 | 7.82 | 0.477 | 100 x 50 x 5 |
| 13.4 | 4.08 | 9.00 | 1.94 | 0.764 | 425 | 10.21 | 65.5 | 1.571 | 4.99 | 1.965 | 1.96 | 0.772 | 68.0 | 4.150 | 14.4 | 0.879 | 125 x 65 x 6 |
| 18.6 | 5.67 | 12.50 | 2.31 | 0.909 | 864 | 20.76 | 122 | 2.931 | 6.04 | 2.378 | 2.27 | 0.894 | 115 | 7.018 | 23.6 | 1.440 | 150 x 75 x 6.5 |
| 24.0 | 7.32 | 16.13 | 2.31 | 0.909 | 1060 | 25.47 | 151 | 3.628 | 5.87 | 2.311 | 2.22 | 0.874 | 141 | 8.604 | 29.1 | 1.776 | 150 x 75 x 9 |
| 21.4 | 6.52 | 14.38 | 2.15 | 0.846 | 1380 | 33.15 | 137 | 3.291 | 7.13 | 2.807 | 2.24 | 0.882 | 154 | 9.398 | 25.5 | 1.556 | 180 x 75 x 7 |
| 27.1 | 8.26 | 18.21 | 2.85 | 1.122 | 1840 | 44.21 | 258 | 6.198 | 7.29 | 2.870 | 2.73 | 1.075 | 204 | 12.45 | 42.0 | 2.563 | 180 x 90 x 7.5 |
| 24.6 | 7.50 | 16.53 | 2.24 | 0.882 | 1950 | 46.85 | 177 | 4.525 | 7.89 | 3.106 | 2.38 | 0.937 | 195 | 11.90 | 30.8 | 1.880 | 200 x 80 x 7.5 |
| 30.3 | 9.24 | 20.36 | 2.77 | 1.091 | 2490 | 59.82 | 286 | 6.871 | 8.03 | 3.161 | 2.72 | 1.071 | 249 | 15.19 | 45.9 | 2.801 | 200 x 90 x 8 |
| 28.4 | 8.66 | 19.08 | 2.15 | 0.846 | 2900 | 69.67 | 200 | 4.805 | 8.96 | 3.528 | 2.35 | 0.925 | 252 | 15.38 | 34.2 | 2.087 | 230 x 80 x 8 |
| 33.1 | 10.09 | 22.24 | 2.58 | 1.016 | 3490 | 83.85 | 303 | 7.280 | 9.10 | 3.583 | 2.68 | 1.055 | 304 | 18.55 | 47.3 | 2.886 | 230 x 90 x 8.5 |
| 30.2 | 9.21 | 20.29 | 2.11 | 0.831 | 3630 | 87.21 | 210 | 5.045 | 9.71 | 3.823 | 2.34 | 0.921 | 291 | 17.76 | 35.7 | 2.179 | 250 x 80 x 8 |
| 34.6 | 10.55 | 23.25 | 2.42 | 0.953 | 4180 | 100.4 | 306 | 7.352 | 9.74 | 3.835 | 2.64 | 1.039 | 335 | 20.44 | 46.5 | 2.838 | 250 x 90 x 9 |
| 40.2 | 12.25 | 27.01 | 2.39 | 0.941 | 4690 | 112.7 | 342 | 8.217 | 9.57 | 3.768 | 2.58 | 1.016 | 375 | 22.88 | 51.7 | 3.155 | 250 x 90 x 11 |
| 38.8 | 11.83 | 26.07 | 2.64 | 1.040 | 5930 | 142.5 | 428 | 10.28 | 11.0 | 4.331 | 2.95 | 1.161 | 423 | 25.81 | 58.2 | 3.552 | 280 x 100 x 9 |
| 48.2 | 14.69 | 32.39 | 2.68 | 1.055 | 7150 | 171.8 | 515 | 12.37 | 10.8 | 4.252 | 2.90 | 1.142 | 510 | 31.12 | 70.4 | 4.296 | 280 x 100 x 11.5 |
| 38.1 | 11.61 | 25.60 | 2.23 | 0.878 | 6440 | 154.7 | 325 | 7.808 | 11.5 | 4.528 | 2.59 | 1.020 | 429 | 26.18 | 48.0 | 2.929 | 300 x 90 x 9 |
| 43.8 | 13.35 | 29.43 | 2.33 | 0.917 | 7400 | 177.8 | 373 | 8.961 | 11.5 | 4.528 | 2.59 | 1.020 | 494 | 30.15 | 56.0 | 3.417 | 300 x 90 x 10 |
| 54.5 | 16.61 | 36.62 | 2.41 | 0.949 | 14500 | 348.4 | 557 | 13.38 | 14.5 | 5.709 | 2.83 | 1.114 | 762 | 46.50 | 73.3 | 4.473 | 380 x 100 x 10.5 |
| 62.0 | 18.90 | 41.66 | 2.29 | 0.902 | 15600 | 374.8 | 584 | 14.03 | 14.1 | 5.551 | 2.72 | 1.071 | 822 | 50.16 | 75.8 | 4.625 | 380 x 100 x 13 |

Note: Calculated based on 1 kg=2.2046 lb, 1 m=3.2808 feet and 1 in = 2.54 cm



# Dimensions and Properties

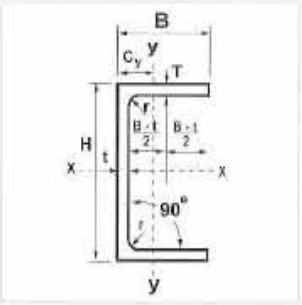



## 5a (iii) Plain Channels

### JIS G 3350:2005 / Manufacturer's Standard

| Designation Size HxB | Unit Weight | Thickness t | Area Of Section A | Centre Of Gravity $C_y$ | Second Moment Of Area $I_x$ | $I_y$ | Radius Of Gyration $r_x$ | $r_y$ | Elastic Modulus $Z_x$ | $Z_y$ | Plastic Modulus $S_x$ | $S_y$ | Buckling Constant u | Torsional Index x | Warping Constant H | Torsional Constant J |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| mm | kg/m | mm | $cm^2$ | cm | $cm^4$ | $cm^4$ | cm | cm | $cm^3$ | $cm^3$ | $cm^3$ | $mm^3$ | | | $dm^2 \times 10^{-2}$ | $cm^4$ |
| 60 x 30 | 1.44 | 1.6 | 1.84 | 0.823 | 10.3 | 1.63 | 2.37 | 0.944 | 3.45 | 0.75 | 4.00 | 1.37 | 0.897 | 29.7 | 0.0101 | 0.0165 |
| | 2.03 | 2.3 | 2.59 | 0.856 | 14.2 | 2.25 | 2.34 | 0.932 | 4.72 | 1.05 | 5.54 | 1.94 | 0.891 | 20.4 | 0.0138 | 0.0490 |
| 75 x 45 | 2.00 | 1.6 | 2.56 | 1.30 | 23.9 | 5.37 | 3.06 | 1.45 | 6.37 | 1.68 | 7.22 | 5.85 | 0.891 | 37.7 | 0.0528 | 0.0226 |
| | 2.84 | 2.3 | 3.62 | 1.34 | 33.1 | 7.49 | 3.02 | 1.44 | 8.83 | 2.37 | 10.1 | 7.83 | 0.885 | 26.1 | 0.0739 | 0.0673 |
| | 3.65 | 3.0 | 4.65 | 1.37 | 41.7 | 9.46 | 2.99 | 1.43 | 11.1 | 3.02 | 12.9 | 9.51 | 0.878 | 19.9 | 0.0939 | 0.150 |
| 100 x 50 | 2.44 | 1.6 | 3.12 | 1.32 | 50.0 | 7.87 | 4.01 | 1.59 | 10.0 | 2.14 | 11.5 | 3.88 | 0.901 | 50.1 | 0.136 | 0.0274 |
| | 3.47 | 2.3 | 4.43 | 1.36 | 69.9 | 11.0 | 3.97 | 1.58 | 14.0 | 3.03 | 16.1 | 5.52 | 0.898 | 34.6 | 0.189 | 0.0815 |
| | 4.47 | 3.0 | 5.70 | 1.39 | 88.6 | 14.0 | 3.94 | 1.57 | 17.7 | 3.88 | 20.6 | 7.12 | 0.895 | 26.3 | 0.240 | 0.181 |
| | 5.87 | 4.0 | 7.47 | 1.43 | 113.3 | 18.0 | 3.89 | 1.55 | 22.7 | 5.05 | 26.6 | 9.37 | 0.890 | 19.5 | 0.307 | 0.430 |
| 125 x 50 | 3.92 | 2.3 | 5.00 | 1.21 | 117 | 11.8 | 4.85 | 1.54 | 18.8 | 3.12 | 22.0 | 5.72 | 0.900 | 43.2 | 0.315 | 0.0916 |
| | 5.06 | 3.0 | 6.45 | 1.24 | 149 | 15.1 | 4.81 | 1.53 | 23.9 | 4.01 | 28.2 | 7.46 | 0.898 | 32.7 | 0.397 | 0.204 |
| | 6.65 | 4.0 | 8.47 | 1.29 | 192 | 19.4 | 4.76 | 1.51 | 30.7 | 5.22 | 36.6 | 9.97 | 0.896 | 24.2 | 0.505 | 0.483 |
| | 7.42 | 4.5 | 9.46 | 1.31 | 212 | 21.4 | 4.74 | 1.51 | 34.0 | 5.81 | 40.6 | 11.2 | 0.895 | 21.3 | 0.555 | 0.689 |
| 150 x 65 | 4.91 | 2.3 | 6.27 | 1.61 | 218 | 25.8 | 5.90 | 2.03 | 29.1 | 5.29 | 33.7 | 9.56 | 0.902 | 52.2 | 1.00 | 0.114 |
| | 6.36 | 3.0 | 8.10 | 1.64 | 279 | 33.1 | 5.87 | 2.02 | 37.2 | 6.81 | 43.4 | 12.4 | 0.900 | 39.7 | 1.27 | 0.253 |
| | 8.38 | 4.0 | 10.67 | 1.69 | 361 | 42.9 | 5.82 | 2.00 | 48.2 | 8.91 | 56.6 | 16.5 | 0.898 | 29.4 | 1.63 | 0.601 |
| | 9.36 | 4.5 | 11.9 | 1.71 | 401 | 47.6 | 5.79 | 2.00 | 53.4 | 9.94 | 63.0 | 18.5 | 0.897 | 26.0 | 1.81 | 0.856 |
| 175 x 75 | 5.73 | 2.3 | 7.30 | 1.83 | 347 | 40.2 | 6.89 | 2.35 | 39.7 | 7.09 | 45.9 | 12.8 | 0.902 | 61.1 | 2.12 | 0.132 |
| | 7.42 | 3.0 | 9.45 | 1.86 | 445 | 51.5 | 6.86 | 2.33 | 50.8 | 9.14 | 59.1 | 16.6 | 0.901 | 46.5 | 2.70 | 0.294 |
| | 9.79 | 4.0 | 12.47 | 1.91 | 579 | 67.1 | 6.81 | 2.32 | 66.1 | 12.0 | 77.4 | 22.1 | 0.899 | 34.6 | 3.49 | 0.697 |
| | 10.95 | 4.5 | 14.0 | 1.93 | 643 | 74.6 | 6.79 | 2.31 | 73.5 | 13.4 | 86.3 | 24.8 | 0.898 | 30.6 | 3.87 | 0.992 |
| 200 x 75 | 6.18 | 2.3 | 7.88 | 1.71 | 473 | 41.7 | 7.75 | 2.30 | 47.3 | 7.21 | 55.4 | 13.0 | 0.899 | 70.0 | 2.88 | 0.142 |
| | 8.01 | 3.0 | 10.2 | 1.74 | 608 | 53.6 | 7.72 | 2.29 | 60.8 | 9.30 | 71.4 | 16.9 | 0.898 | 53.3 | 3.67 | 0.316 |
| | 10.6 | 4.0 | 13.5 | 1.78 | 792 | 69.8 | 7.67 | 2.28 | 79.2 | 12.2 | 94 | 22.7 | 0.897 | 39.5 | 4.74 | 0.75 |
| | 11.84 | 4.5 | 15.1 | 1.80 | 881 | 77.7 | 7.64 | 2.27 | 88.1 | 13.6 | 104 | 25.6 | 0.897 | 35.0 | 5.24 | 1.07 |
| 225 x 75 | 6.63 | 2.3 | 8.45 | 1.60 | 624 | 43.1 | 8.60 | 2.26 | 55.5 | 7.30 | 65.6 | 13.2 | 0.894 | 79.0 | 3.79 | 0.152 |
| | 8.59 | 3.0 | 11.0 | 1.63 | 803 | 55.4 | 8.56 | 2.25 | 71.3 | 9.43 | 84.6 | 17.3 | 0.894 | 60.1 | 4.83 | 0.339 |
| | 11.40 | 4.0 | 14.5 | 1.67 | 1048 | 72.2 | 8.51 | 2.23 | 93 | 12.4 | 111 | 23.3 | 0.893 | 44.6 | 6.22 | 0.80 |
| | 12.72 | 4.5 | 16.2 | 1.70 | 1166 | 80.4 | 8.48 | 2.23 | 104 | 13.8 | 124 | 26.3 | 0.893 | 39.4 | 6.88 | 1.14 |
| 250 x 75 | 7.08 | 2.3 | 9.03 | 1.50 | 802 | 44.3 | 9.43 | 2.22 | 64.2 | 7.39 | 76.5 | 13.4 | 0.888 | 88.2 | 4.84 | 0.163 |
| | 9.18 | 3.0 | 11.7 | 1.54 | 1032 | 56.9 | 9.39 | 2.21 | 82.5 | 9.54 | 98.8 | 17.6 | 0.888 | 67.1 | 6.17 | 0.361 |
| | 12.10 | 4.0 | 15.5 | 1.58 | 1349 | 74.3 | 9.34 | 2.19 | 108 | 12.5 | 130 | 23.9 | 0.888 | 49.8 | 7.95 | 0.86 |
| | 13.60 | 4.5 | 17.3 | 1.60 | 1502 | 82.7 | 9.31 | 2.18 | 120 | 14.0 | 145 | 27.1 | 0.888 | 44.0 | 8.79 | 1.22 |

# Dimensions and Properties




## 5a (iv) Parallel Flanges

### Part 1

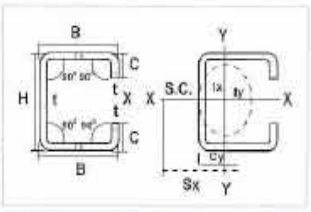
| Designation Size H x B | Unit Weight | Thickness Web t | Thickness Flange T | Root Radius r | Depth Between Fillets d | Area Of Section A | Centre Of Gravity $C_y$ | Ratios For Local Buckling Flange B/T | Ratios For Local Buckling Web d/t | Second Moment Of Area Axis x-x | Second Moment Of Area Axis y-y |
|---|---|---|---|---|---|---|---|---|---|---|---|
| mm x mm | kg/m | mm | mm | mm | mm | $cm^2$ | cm | | | $cm^4$ | $cm^4$ |
| 100 x 50 | 10.2 | 5.0 | 8.5 | 9 | 65.0 | 13.0 | 1.73 | 5.88 | 13.0 | 208 | 32.3 |
| 125 x 65 | 14.8 | 5.5 | 9.5 | 12 | 82.0 | 18.8 | 2.25 | 6.84 | 14.9 | 483 | 80.0 |
| 150 x 75 | 17.9 | 5.5 | 10.0 | 12 | 106 | 22.8 | 2.58 | 7.50 | 19.3 | 861 | 131 |
| 150 x 90 | 23.9 | 6.5 | 12.0 | 12 | 102 | 30.4 | 3.30 | 7.50 | 15.7 | 1162 | 253 |
| 180 x 75 | 20.3 | 6.0 | 10.5 | 12 | 135 | 25.9 | 2.41 | 7.14 | 22.5 | 1370 | 146 |
| 180 x 90 | 26.1 | 6.5 | 12.5 | 12 | 131 | 33.2 | 3.17 | 7.20 | 20.2 | 1817 | 277 |
| 200 x 75 | 23.4 | 6.0 | 12.5 | 12 | 151 | 29.9 | 2.48 | 6.00 | 25.2 | 1963 | 170 |
| 200 x 90 | 29.7 | 7.0 | 14.0 | 12 | 148 | 37.9 | 3.12 | 6.43 | 21.1 | 2523 | 314 |
| 230 x 75 | 25.7 | 6.5 | 12.5 | 12 | 181 | 32.7 | 2.30 | 6.00 | 27.8 | 2748 | 181 |
| 230 x 90 | 32.2 | 7.5 | 14.0 | 12 | 178 | 41.0 | 2.92 | 6.43 | 23.7 | 3518 | 334 |
| 250 x 90 | 35.5 | 8.0 | 15.0 | 12 | 220 | 45.2 | 2.86 | 6.00 | 27.5 | 4510 | 364 |
| 260 x 75 | 27.6 | 7.0 | 12.0 | 12 | 212 | 35.1 | 2.10 | 6.25 | 30.3 | 3619 | 185 |
| 260 x 90 | 34.8 | 8.0 | 14.0 | 12 | 208 | 44.4 | 2.74 | 6.43 | 26.0 | 4728 | 353 |
| 300 x 90 | 41.4 | 9.0 | 15.5 | 12 | 245 | 52.7 | 2.60 | 5.81 | 27.2 | 7218 | 404 |
| 300 x 100 | 45.5 | 9.0 | 16.5 | 15 | 237 | 58.0 | 3.05 | 6.06 | 26.3 | 8229 | 568 |
| 380 x 100 | 54.0 | 9.5 | 17.5 | 15 | 315 | 68.7 | 2.79 | 5.71 | 33.2 | 15030 | 643 |
| 430 x 100 | 64.4 | 11.0 | 19.0 | 15 | 362 | 82.1 | 2.62 | 5.26 | 32.9 | 21940 | 722 |

Note : The flange thickness is measured at the centre of the flange

### Part 2

| Designation Size H x B | Unit Weight | Radius Of Gyration Axis x-x | Radius Of Gyration Axis y-y | Elastic Modulus Axis x-x | Elastic Modulus Axis y-y | Plastic Modulus Axis x-x | Plastic Modulus Axis y-y | Buckling Parameter | Torsional Index | Warping Constant | Torsional Constant |
|---|---|---|---|---|---|---|---|---|---|---|---|
| mm x mm | kg/m | cm | cm | $cm^3$ | $cm^3$ | $cm^3$ | $cm^3$ | | | $dm^6$ | $cm^4$ |
| 100 x 50 | 10.2 | 4.00 | 1.58 | 41.5 | 9.89 | 48.9 | 17.5 | 0.942 | 10.0 | 0.000491 | 2.53 |
| 125 x 65 | 14.8 | 5.07 | 2.06 | 77.3 | 18.8 | 89.9 | 33.2 | 0.942 | 11.1 | 0.00194 | 4.72 |
| 150 x 75 | 17.9 | 6.15 | 2.40 | 115 | 26.6 | 132 | 47.2 | 0.946 | 13.1 | 0.00467 | 6.10 |
| 150 x 90 | 23.9 | 6.18 | 2.89 | 155 | 44.4 | 179 | 76.9 | 0.936 | 10.8 | 0.00890 | 11.8 |
| 180 x 75 | 20.3 | 7.27 | 2.38 | 152 | 28.8 | 176 | 51.8 | 0.946 | 15.3 | 0.00754 | 7.34 |
| 180 x 90 | 26.1 | 7.40 | 2.89 | 202 | 47.4 | 232 | 83.5 | 0.949 | 12.8 | 0.0141 | 13.3 |
| 200 x 75 | 23.4 | 8.11 | 2.39 | 196 | 33.8 | 227 | 60.6 | 0.956 | 14.8 | 0.0107 | 11.1 |
| 200 x 90 | 29.7 | 8.16 | 2.89 | 252 | 53.4 | 291 | 94.5 | 0.954 | 12.9 | 0.0197 | 18.3 |
| 230 x 75 | 25.7 | 9.17 | 2.35 | 239 | 34.8 | 278 | 63.2 | 0.947 | 17.3 | 0.0153 | 11.8 |
| 230 x 90 | 32.2 | 9.27 | 2.86 | 306 | 55.0 | 355 | 98.9 | 0.950 | 15.1 | 0.0279 | 19.3 |
| 250 x 90 | 35.5 | 9.99 | 2.84 | 361 | 59.3 | 421 | 107 | 0.948 | 15.5 | 0.0359 | 23.8 |
| 260 x 75 | 27.6 | 10.1 | 2.30 | 278 | 34.4 | 328 | 62.0 | 0.932 | 20.5 | 0.0203 | 11.7 |
| 260 x 90 | 34.8 | 10.3 | 2.82 | 364 | 56.3 | 425 | 102 | 0.942 | 17.2 | 0.0379 | 20.6 |
| 300 x 90 | 41.4 | 11.7 | 2.77 | 481 | 63.1 | 568 | 114 | 0.934 | 18.4 | 0.0581 | 28.8 |
| 300 x 100 | 45.5 | 11.9 | 3.13 | 549 | 81.7 | 641 | 148 | 0.944 | 17.0 | 0.0813 | 36.8 |
| 380 x 100 | 54.0 | 14.8 | 3.06 | 791 | 89.2 | 933 | 161 | 0.932 | 21.2 | 0.150 | 45.7 |
| 430 x 100 | 64.4 | 16.3 | 2.97 | 1020 | 97.9 | 1222 | 176 | 0.917 | 22.5 | 0.219 | 63.0 |

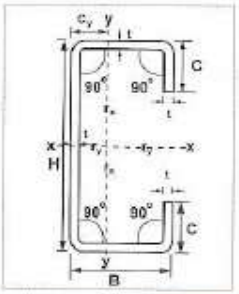





## 5b (i) Trolley Track

| Dimension H x B x C mm (in) | Thickness t mm | Section Area A $cm^2$ | Unit Weight M Kg/m | Centre of Gravity $C_x$ cm | Centre of Gravity $C_y$ cm | Secondary Moment of Area $I_x$ $cm^4$ | Secondary Moment of Area $I_y$ $cm^4$ | Radius of Gyration of Area $i_x$ cm | Radius of Gyration of Area $i_y$ cm | Modulus of Section $Z_x$ $cm^3$ | Modulus of Section $Z_y$ $cm^3$ | Centre of Shear $S_x$ cm | Centre of Shear $S_y$ cm |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 50 x 50 x 15 | 1.5 | 2.55 | 2.00 | 2.50 | 2.22 | 11.32 | 9.45 | 2.08 | 1.90 | 4.53 | 3.40 | 4.7 | 0 |
| (2 x 2 x 5/8) | 2.3 | 3.79 | 2.98 | 2.50 | 2.22 | 16.53 | 13.76 | 2.05 | 1.87 | 6.61 | 4.95 | 4.7 | 0 |

## Dimensions and Properties






# 5b (ii) Lip Channels
# JIS G 3350:2005 / Manufacturer's Standard

| Designation | | Thickness | Inside Outside Radius | | Area Of Section | Centre Of Gravity | Second Moment Of Area | | Radius Of Gyration | | Elastic Modulus | | Plastic Modulus | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Size H x B x C | Unit Weight | t | $r_1$ | $r_2$ | A | $C_y$ | $I_x$ | $I_y$ | $r_x$ | $r_y$ | $Z_x$ | $Z_y$ | $S_x$ | $S_y$ |
| mm ( in ) | kg/m | mm | mm | mm | $cm^2$ | cm | $cm^4$ | $cm^4$ | cm | cm | $cm^3$ | $cm^3$ | $cm^3$ | $cm^3$ |
| 60 x 30 x 10 | 1.62 | 1.6 | 4.0 | 2.4 | 2.05 | 1.06 | 11.4 | 2.50 | 2.36 | 1.10 | 3.81 | 1.29 | 4.48 | 2.12 |
| ( 2 1/2 x 1 1/4 x 3/8 ) | 1.99 | 2.0 | 5.0 | 3.0 | 2.50 | 1.05 | 13.7 | 2.93 | 2.34 | 1.08 | 4.56 | 1.50 | 5.41 | 2.54 |
| | 2.25 | 2.3 | 5.8 | 3.5 | 2.83 | 1.05 | 15.2 | 3.20 | 2.32 | 1.06 | 5.06 | 1.64 | 6.05 | 2.57 |
| 75 x 45 x 15 | 2.31 | 1.6 | 4.0 | 2.4 | 2.93 | 1.72 | 26.8 | 8.59 | 3.03 | 1.71 | 7.15 | 3.09 | 8.27 | 4.62 |
| ( 3 x 1 3/4 x 1/2 ) | 2.86 | 2.0 | 5.0 | 3.0 | 3.60 | 1.71 | 32.5 | 10.3 | 3.00 | 1.69 | 8.66 | 3.69 | 10.1 | 5.61 |
| | 3.24 | 2.3 | 5.8 | 3.5 | 4.09 | 1.71 | 36.5 | 11.5 | 2.99 | 1.67 | 9.73 | 4.11 | 11.4 | 6.31 |
| | 4.12 | 3.0 | 7.5 | 4.5 | 5.18 | 1.70 | 44.9 | 13.8 | 2.94 | 1.63 | 12.0 | 4.94 | 14.2 | 7.79 |
| | 4.37 | 3.2 | 8.0 | 4.8 | 5.48 | 1.70 | 47.1 | 14.4 | 2.93 | 1.62 | 12.6 | 5.15 | 14.9 | 8.18 |
| 100 x 50 x 20 | 2.88 | 1.6 | 4.0 | 2.4 | 3.65 | 1.86 | 57.9 | 13.8 | 3.98 | 1.95 | 11.6 | 4.41 | 13.5 | 6.73 |
| (4 x 2 x 3/4 ) | 3.60 | 2.0 | 5.0 | 3.0 | 4.50 | 1.86 | 70.5 | 16.7 | 3.96 | 1.93 | 14.1 | 5.32 | 16.5 | 8.22 |
| | 4.06 | 2.3 | 5.8 | 3.5 | 5.13 | 1.86 | 79.6 | 18.7 | 3.94 | 1.91 | 15.9 | 5.95 | 18.7 | 9.28 |
| | 5.18 | 3.0 | 7.5 | 4.5 | 6.53 | 1.85 | 99 | 22.9 | 3.90 | 1.87 | 19.8 | 7.27 | 23.6 | 11.6 |
| | 5.50 | 3.2 | 8.0 | 4.8 | 6.92 | 1.85 | 104 | 24.0 | 3.88 | 1.86 | 20.9 | 7.60 | 24.9 | 12.2 |
| | 6.71 | 4.0 | 10.0 | 6.0 | 8.41 | 1.84 | 123 | 27.7 | 3.83 | 1.82 | 24.7 | 8.79 | 29.8 | 14.5 |
| | 7.43 | 4.5 | 11.3 | 6.8 | 9.29 | 1.84 | 134 | 29.7 | 3.80 | 1.79 | 26.8 | 9.40 | 32.6 | 14.4 |
| 125 x 50 x 20 | 3.95 | 2.0 | 5.0 | 3.0 | 5.00 | 1.68 | 119 | 18.1 | 4.88 | 1.90 | 19.1 | 5.46 | 22.5 | 7.90 |
| ( 5 x 2 x 3/4 ) | 4.51 | 2.3 | 5.8 | 3.5 | 5.70 | 1.68 | 135 | 20.3 | 4.86 | 1.89 | 21.5 | 6.11 | 25.5 | 8.91 |
| | 5.77 | 3.0 | 7.5 | 4.5 | 7.28 | 1.68 | 169 | 24.8 | 4.81 | 1.85 | 27.0 | 7.47 | 32.2 | 11.2 |
| | 6.13 | 3.2 | 8.0 | 4.8 | 7.72 | 1.68 | 178 | 26.0 | 4.80 | 1.84 | 28.4 | 7.82 | 34.0 | 11.8 |
| | 7.50 | 4.0 | 10.0 | 6.0 | 9.41 | 1.67 | 212 | 30.2 | 4.74 | 1.79 | 33.9 | 9.05 | 40.9 | 14.1 |
| | 8.31 | 4.5 | 11.3 | 6.8 | 10.4 | 1.66 | 231 | 32.3 | 4.71 | 1.76 | 36.9 | 9.70 | 45.0 | 15.4 |
| 150 x 65 x 20 | 5.50 | 2.3 | 5.8 | 3.5 | 6.97 | 2.11 | 245 | 40.5 | 5.93 | 2.41 | 32.7 | 9.23 | 38.1 | 13.8 |
| ( 6 x 2 1/2 x 3/4 ) | 7.07 | 3.0 | 7.5 | 4.5 | 8.93 | 2.10 | 310 | 50.2 | 5.89 | 2.37 | 41.3 | 11.4 | 48.4 | 17.4 |
| | 7.51 | 3.2 | 8.0 | 4.8 | 9.48 | 2.10 | 327 | 52.7 | 5.87 | 2.36 | 43.6 | 12.0 | 51.2 | 18.4 |
| | 9.22 | 4.0 | 10.0 | 6.0 | 11.6 | 2.09 | 393 | 62.0 | 5.82 | 2.31 | 52.4 | 14.1 | 62.1 | 22.1 |
| | 10.25 | 4.5 | 11.3 | 6.8 | 12.9 | 2.09 | 432 | 67.1 | 5.78 | 2.28 | 57.5 | 15.2 | 68.5 | 24.2 |
| 175 x 75 x 20 | 6.31 | 2.3 | 5.8 | 3.5 | 8.00 | 2.34 | 386 | 60.3 | 6.95 | 2.75 | 44.1 | 11.7 | 51.1 | 17.8 |
| ( 7 x 3 x 3/4 ) | 8.13 | 3.0 | 7.5 | 4.5 | 10.3 | 2.33 | 489 | 75.1 | 6.90 | 2.70 | 55.9 | 14.5 | 65.2 | 22.5 |
| | 8.63 | 3.2 | 8.0 | 4.8 | 10.9 | 2.33 | 518 | 79.1 | 6.88 | 2.69 | 59.1 | 15.3 | 69.1 | 23.7 |
| | 10.63 | 4.0 | 10.0 | 6.0 | 13.4 | 2.32 | 626 | 93.6 | 6.83 | 2.64 | 71.5 | 18.1 | 84.1 | 28.7 |
| | 11.84 | 4.5 | 11.3 | 6.8 | 14.9 | 2.31 | 689 | 102 | 6.80 | 2.61 | 78.7 | 19.6 | 93.0 | 31.6 |
| 200 x 75 x 20 | 6.76 | 2.3 | 5.8 | 3.5 | 8.58 | 2.19 | 527 | 63.0 | 7.84 | 2.71 | 52.7 | 11.9 | 61.5 | 18.0 |
| ( 10 x 3 x 3/4 ) | 8.71 | 3.0 | 7.5 | 4.5 | 11.0 | 2.18 | 669 | 78.5 | 7.79 | 2.67 | 66.9 | 14.8 | 78.5 | 22.9 |
| | 9.27 | 3.2 | 8.0 | 4.8 | 11.7 | 2.18 | 708 | 82.6 | 7.77 | 2.65 | 70.8 | 15.5 | 83.2 | 24.3 |
| | 11.40 | 4.0 | 10.0 | 6.0 | 14.4 | 2.17 | 857 | 98 | 7.71 | 2.61 | 85.7 | 18.4 | 101 | 29.5 |
| | 12.73 | 4.5 | 11.3 | 6.8 | 16.0 | 2.17 | 945 | 106 | 7.68 | 2.57 | 94.5 | 19.9 | 112 | 32.6 |
| 200 x 75 x 25 | 6.95 | 2.3 | 5.8 | 3.5 | 8.81 | 2.33 | 540 | 69.0 | 7.83 | 2.80 | 54.0 | 13.3 | 63.3 | 19.6 |
| ( 10 x 3 x 1 ) | 8.93 | 3.0 | 7.5 | 4.5 | 11.3 | 2.32 | 687 | 86.3 | 7.78 | 2.76 | 68.7 | 16.7 | 80.8 | 24.9 |
| | 9.52 | 3.2 | 8.0 | 4.8 | 12.0 | 2.32 | 727 | 90.9 | 7.77 | 2.75 | 72.7 | 17.5 | 85.7 | 26.4 |
| | 11.70 | 4.0 | 10.0 | 6.0 | 14.8 | 2.31 | 881 | 108 | 7.71 | 2.70 | 88.1 | 20.8 | 105 | 32.0 |
| | 13.10 | 4.5 | 11.3 | 6.8 | 16.5 | 2.31 | 972 | 118 | 7.68 | 2.67 | 97.2 | 22.7 | 116 | 35.4 |
| 225 x 75 x 20 | 7.21 | 2.3 | 5.8 | 3.5 | 9.15 | 2.06 | 694 | 65.3 | 8.71 | 2.67 | 61.7 | 12.0 | 72.6 | 18.3 |
| ( 9 x 3 x 3/4 ) | 9.30 | 3.0 | 7.5 | 4.5 | 11.8 | 2.05 | 883 | 81.4 | 8.66 | 2.63 | 78.4 | 14.9 | 92.8 | 23.4 |
| | 9.89 | 3.2 | 8.0 | 4.8 | 12.5 | 2.05 | 934 | 85.7 | 8.64 | 2.62 | 83.1 | 15.7 | 98.4 | 24.8 |
| | 12.20 | 4.0 | 10.0 | 6.0 | 15.4 | 2.05 | 1134 | 101 | 8.58 | 2.57 | 101 | 18.6 | 120 | 30.3 |
| | 13.61 | 4.5 | 11.3 | 6.8 | 17.2 | 2.04 | 1252 | 110 | 8.54 | 2.53 | 111 | 20.2 | 133 | 33.6 |
| 225 x 75 x 25 | 7.40 | 2.3 | 5.8 | 3.5 | 9.38 | 2.19 | 713 | 71.6 | 8.72 | 2.76 | 63.3 | 13.5 | 74.6 | 19.9 |
| ( 9 x 3 x 1 ) | 9.54 | 3.0 | 7.5 | 4.5 | 12.1 | 2.19 | 907 | 89.6 | 8.66 | 2.72 | 80.6 | 16.9 | 95.5 | 25.3 |
| | 10.10 | 3.2 | 8.0 | 4.8 | 12.8 | 2.18 | 960 | 94.4 | 8.65 | 2.71 | 85.4 | 17.8 | 101 | 26.9 |
| | 12.50 | 4.0 | 10.0 | 6.0 | 15.8 | 2.18 | 1166 | 112 | 8.59 | 2.66 | 104 | 21.1 | 124 | 32.8 |
| | 14.00 | 4.5 | 11.3 | 6.8 | 17.6 | 2.17 | 1288 | 122 | 8.55 | 2.64 | 115 | 23.0 | 137 | 36.4 |
| 250 x 75 x 20 | 7.67 | 2.3 | 5.8 | 3.5 | 9.73 | 1.94 | 890 | 67.3 | 9.57 | 2.63 | 71.2 | 12.1 | 84.4 | 18.6 |
| ( 10 x 3 x 3/4 ) | 9.89 | 3.0 | 7.5 | 4.5 | 12.5 | 1.94 | 1133 | 83.9 | 9.51 | 2.59 | 90.7 | 15.1 | 108 | 23.8 |
| | 10.52 | 3.2 | 8.0 | 4.8 | 13.3 | 1.94 | 1200 | 88.4 | 9.49 | 2.58 | 96.0 | 15.9 | 115 | 25.3 |
| | 12.99 | 4.0 | 10.0 | 6.0 | 16.4 | 1.93 | 1459 | 105 | 9.43 | 2.53 | 117 | 18.8 | 140 | 31.1 |
| | 14.49 | 4.5 | 11.3 | 6.8 | 18.3 | 1.93 | 1612 | 114 | 9.39 | 2.49 | 129 | 20.4 | 155 | 34.6 |
| 250 x 75 x 25 | 7.85 | 2.3 | 5.8 | 3.5 | 10.0 | 2.07 | 914 | 74.0 | 9.58 | 2.73 | 73.1 | 13.6 | 86.7 | 20.1 |
| ( 10 x 3 x 1 ) | 10.13 | 3.0 | 7.5 | 4.5 | 12.8 | 2.07 | 1165 | 92.5 | 9.53 | 2.69 | 93.2 | 17.0 | 111 | 25.8 |
| | 10.80 | 3.2 | 8.0 | 4.8 | 13.6 | 2.07 | 1234 | 97.5 | 9.51 | 2.67 | 98.7 | 17.9 | 118 | 27.4 |
| | 13.30 | 4.0 | 10.0 | 6.0 | 16.8 | 2.06 | 1501 | 116 | 9.45 | 2.63 | 120 | 21.3 | 144 | 33.6 |
| | 14.85 | 4.5 | 11.3 | 6.8 | 18.7 | 2.06 | 1659 | 126 | 9.41 | 2.60 | 133 | 23.2 | 160 | 37.4 |